

Databases Laboratory 1



Marin Iuliana
marin.iulliana25@gmail.com

Lab Objectives



- > Create your Dev/Test Environment with Oracle Application Express or SQL Developer
- > Retrieve Data Using the SQL SELECT Statement

Login to Oracle Application Express



- Open a browser and access the following website (<https://iacademy3.oracle.com/>) with your account details from the SQL_Accounts_DB.xlsx file. Password: gPqKX2o*

A screenshot of the Oracle Application Express login page. At the top, there is a logo consisting of two crossed pencils, one yellow and one blue. Below the logo, the text "Oracle Application Express" is displayed. There are three input fields, each with a green checkmark on the right side, indicating successful validation. The first field has a database icon and contains the text "RO_A398_SQL_S01". The second field has a user icon and contains the text "RO_A398_SQL_S01". The third field has a key icon and contains a series of dots representing a password. Below these fields is a large blue button labeled "Sign In". At the bottom, there is a link labeled "Reset Password".

Oracle Application Express

RO_A398_SQL_S01 ✓

RO_A398_SQL_S01 ✓

..... ✓

Sign In

[Reset Password](#)

How to add tables and data to the APEX account



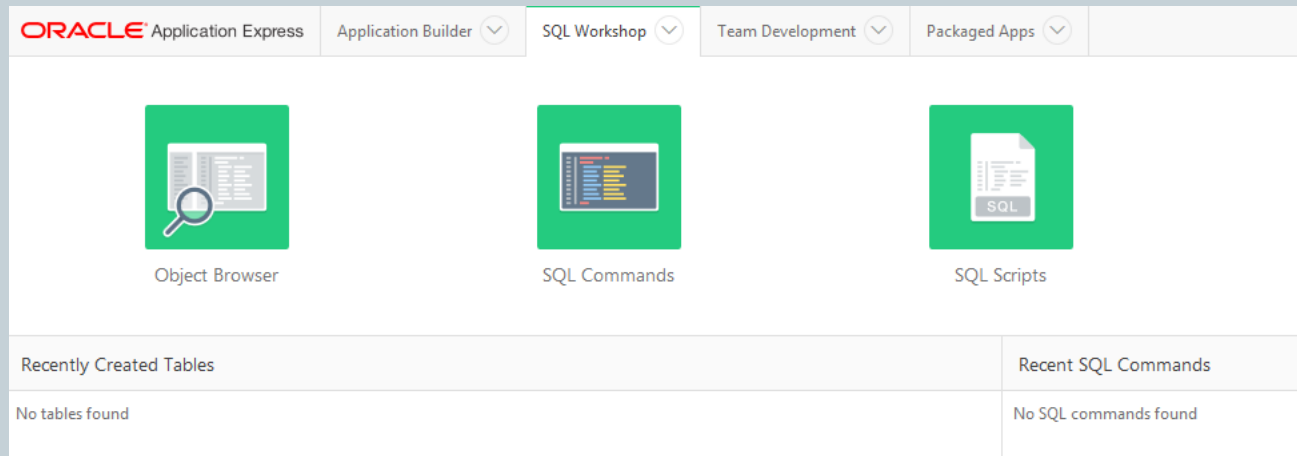
- Download the SQL_Schema.sql script file.
- Open APEX in your browser and login.
- Select “SQL Workshop”

The screenshot displays the Oracle APEX interface. At the top, there is a navigation bar with the following items: "ORACLE Application Express", "Application Builder" (with a dropdown arrow), "SQL Workshop" (with a dropdown arrow and highlighted), "Team Development" (with a dropdown arrow), and "Packaged Apps" (with a dropdown arrow). Below the navigation bar, there are three main icons: "Application Builder" (blue square with a pencil and grid), "SQL Workshop" (green square with a code editor and upload icon), and "Team Development" (yellow square with a circular flow diagram). Below these icons, there are three sections: "Top Applications", "Top Users", and "News and Me". The "Top Users" section shows a table with one user: "us_z101_sql_s01" with a value of "1".

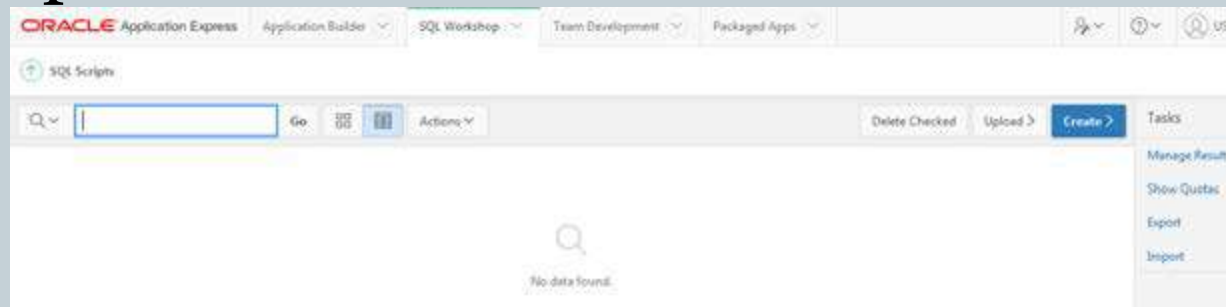
Top Applications	Top Users	News and Me						
	<table border="1"><thead><tr><th>US</th><th>us_z101_sql_s01</th><th>1</th></tr></thead><tbody><tr><td colspan="3"> </td></tr></tbody></table>	US	us_z101_sql_s01	1				
US	us_z101_sql_s01	1						

How to add tables and data to the APEX account

- Select “SQL Scripts”



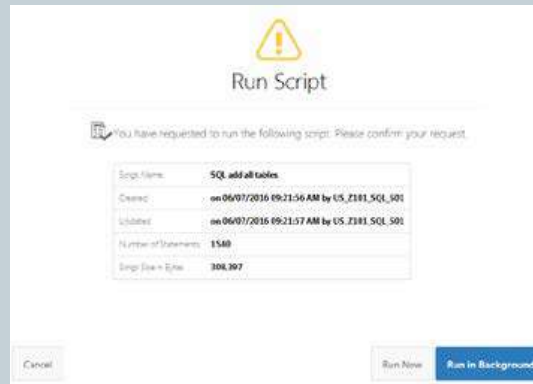
- Click “Upload”



How to add tables and data to the APEX account



- Click Browse and navigate to the .sql file you downloaded.
- Add a Script Name - “SQL add all tables”, leave “File Character Set” as default (Unicode UTF-8), and click “Upload”.
- You will now see the Script listed.
- Click the Run icon.
- Click “Run Now”.



How to add tables and data to the APEX account



- This will take you to the Manage Script Results Page.
 - Click "View Results"
 - The first attempt will generate errors for the DROP statements due to the tables not already existing in the schema. These statements are necessary in the script, if it becomes necessary to restore the tables to the original state.
 - Click the "SQL Workshop" tab
 - Click "Object Browser" .
- 
- The screenshot shows the Oracle SQL Developer interface. On the left, the 'Object Browser' pane is open, displaying a tree view of the database schema. The tree is expanded to show the 'TABLES' folder, which contains a list of tables. A red arrow points from the text 'You should now see the tables listed on the left of the Object Browser page' to this list of tables. The main workspace on the right is currently empty.
- You should now see the tables listed on the left of the Object Browser page. These are the tables (and data) that will be used in the curriculum for your course.

Tables that can be used



The screenshot displays a database management interface. On the left, a sidebar lists various tables, with 'COUNTRIES' selected and highlighted in blue. The main panel shows the 'COUNTRIES' table structure, including tabs for Table, Data, Indexes, Model, Constraints, Grants, Statistics, UI Defaults, Triggers, Dependencies, and SQL. Below these tabs are buttons for 'Add Column', 'Modify Column', 'Rename Column', 'Drop Column', 'Rename', 'Copy', 'Drop', 'Truncate', and 'Create Lookup Table'. The table structure is presented in a table format with columns: Column Name, Data Type, Nullable, Default, and Primary Key. The columns listed are COUNTRY_ID (CHAR(2), No, -, 1), COUNTRY_NAME (VARCHAR2(40), Yes, -, -), and REGION_ID (NUMBER, Yes, -, -). A 'Download' link is visible below the table structure.

Column Name	Data Type	Nullable	Default	Primary Key
COUNTRY_ID	CHAR(2)	No	-	1
COUNTRY_NAME	VARCHAR2(40)	Yes	-	-
REGION_ID	NUMBER	Yes	-	-

Running a query



ORACLE Application Express Application Builder SQL Workshop Team Development Packaged Apps

SQL Commands Schema RO_B720_SQL_T01

Rows 10 Clear Command Find Tables Save Run

```
SELECT DISTINCT job_id
FROM employees;
```

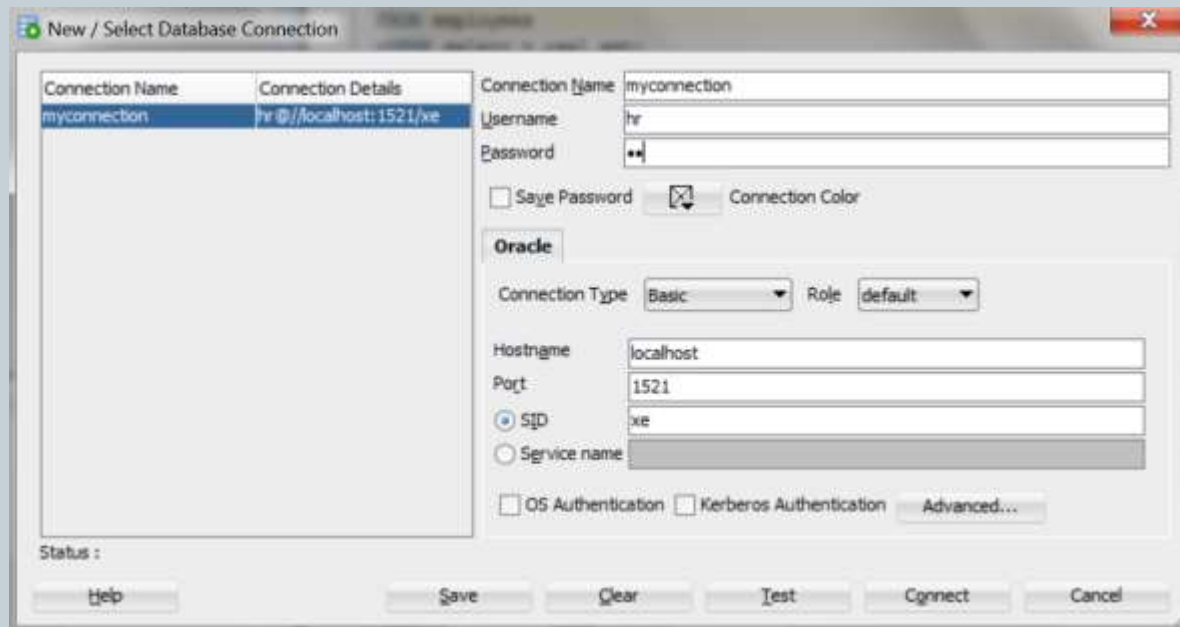
Results Explain Describe Saved SQL History

JOB_ID
AC_ACCOUNT
AC_MGR
AD_ASST
AD PRES
AD_VP
IT_PROG
MK_MAN
MK_REP
SA_MAN

Setting Oracle SQL Developer



- Install Oracle Database 11g Express Edition and Oracle SQL Developer.
- Create a new connection.



Oracle SQL Developer Connection



- Unlock the hr username.

```
C:\Users\Iuliana>sqlplus/nolog
SQL*Plus: Release 11.2.0.2.0 Production on Mon Aug 29 18:08:09 2016
Copyright (c) 1982, 2014, Oracle. All rights reserved.

SQL> connect sys as sysdba
Enter password:
Connected.
SQL> grant create session to ora1;

Grant succeeded.

SQL> alter user hr identified by hr account unlock;

User altered.

SQL>
```

↑
password

SQL Syntax



- **SELECT** column_name, aggregate_function(column_name)
FROM table_name
WHERE column_name operator value
GROUP BY column_name
HAVING aggregate_function(column_name) operator value;

Structure of table EMPLOYEES



```
DESCRIBE employees
Name                Null      Type
-----
EMPLOYEE_ID         NOT NULL  NUMBER(6)
FIRST_NAME
LAST_NAME           NOT NULL  VARCHAR2(25)
EMAIL               NOT NULL  VARCHAR2(25)
PHONE_NUMBER
HIRE_DATE           NOT NULL  DATE
JOB_ID              NOT NULL  VARCHAR2(10)
SALARY
COMMISSION_PCT
MANAGER_ID           NUMBER(6)
DEPARTMENT_ID        NUMBER(4)
```

Queries. Retrieving Data Using the SQL SELECT Statement



- The HR department wants a query to display all unique job IDs from the EMPLOYEES table.

```
DESCRIBE employees
```

```
Name
```

```
-----
```

```
EMPLOYEE_ID
```

```
FIRST_NAME
```

```
LAST_NAME
```

```
EMAIL
```

```
PHONE_NUMBER
```

```
HIRE_DATE
```

```
JOB_ID
```

```
SALARY
```

```
COMMISSION_PCT
```

```
MANAGER_ID
```

```
DEPARTMENT_ID
```

```
SELECT DISTINCT job_id
```

```
FROM employees;
```

The HR department wants a query to display the last name, job ID, hire date, and employee ID for each employee, with the employee ID appearing first.

Provide an alias STARTDATE for the HIRE_DATE column.

```
SELECT employee_id, last_name, job_id, hire_date StartDate  
FROM employees;
```

If you want to name the column **Start Date** instead of the single word StartDate, then use double quotes.

```
SELECT employee_id, last_name, job_id, hire_date "Start Date"  
FROM employees;
```

Queries. Retrieving Data Using the SQL SELECT Statement



- The HR department has requested a report of all employees and their job IDs. Display the last name concatenated with the job ID (separated by a comma and space ‘, ‘) and name the column “Employee and Title”.

```
DESCRIBE employees
Name
-----
EMPLOYEE_ID
FIRST_NAME
LAST_NAME
EMAIL
PHONE_NUMBER
HIRE_DATE
JOB_ID
SALARY
COMMISSION_PCT
MANAGER_ID
DEPARTMENT_ID
```

```
SELECT last_name||', '||job_id "Employee and Title"
FROM employees;
```

**Use || and simple
quotes ‘, ‘ for
concatenation**

Activities



- 1. Does the following SELECT statement execute successfully?

```
SELECT last_name, job_id, salary AS Sal  
FROM employees;
```

- 2. There are four coding errors in the following statement. Can you identify them?

```
SELECT employee_id, last_name  
sal x 12 ANNUAL SALARY  
FROM employees;
```


Activities



- 3. Determine the structure of the EMPLOYEES table.
- 4. Create a query to display all the data from the EMPLOYEES table. Separate each column output by a comma. Name the column THE_OUTPUT.